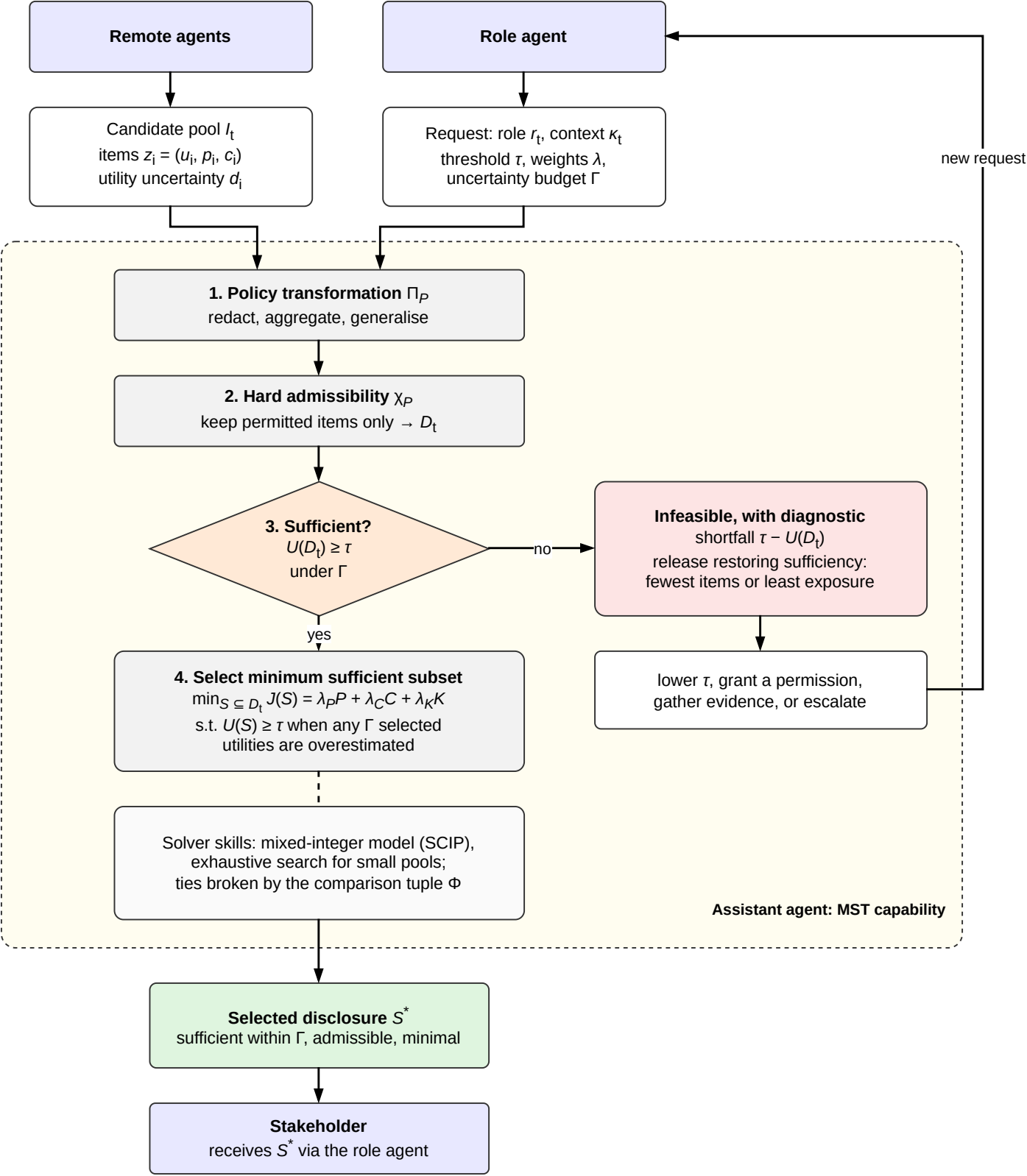
Remote agents
Role agent
Candidate pool $I_t$
items $z_i = (u_i, p_i, c_i)$
utility uncertainty $d_i$
Request: role $r_t$, context $\kappa_t$
threshold $\tau$, weights $\lambda$,
uncertainty budget $\Gamma$
new request
1. Policy transformation $\Pi_P$
redact, aggregate, generalise
2. Hard admissibility $\chi_P$
keep permitted items only → $D_t$
3. Sufficient?
$U(D_t) \geq \tau$
under $\Gamma$
no
Infeasible, with diagnostic
shortfall $\tau - U(D_t)$
release restoring sufficiency:
fewest items or least exposure
yes
lower $\tau$, grant a permission,
gather evidence, or escalate
4. Select minimum sufficient subset
$\min_{S \subseteq D_t} J(S) = \lambda_P P + \lambda_C C + \lambda_K K$
s.t. $U(S) \geq \tau$ when any $\Gamma$ selected
utilities are overestimated
Solver skills: mixed-integer model (SCIP),
exhaustive search for small pools;
ties broken by the comparison tuple $\Phi$
Assistant agent: MST capability
Selected disclosure $S^*$
sufficient within $\Gamma$, admissible, minimal
Stakeholder
receives $S^*$ via the role agent